

Section	Text
Framework title	Circular Window Component and Rebound-Risk Check
Framework purpose	The preliminary framework translates the literature review, VELFAC case study, expert interviews, lifecycle scenario comparison and classified risk pathways into a practical screening tool. Its purpose is to help architects and specifiers assess whether a circular window strategy is likely to deliver real product-life extension or whether its expected benefit may be weakened by implementation gaps, trade-offs, burden shifting, lock-in or rebound-related behaviour.
Why the framework is needed	The project shows that repairability cannot be judged only from the presence of replaceable parts. A window component may be technically replaceable, but repair may still fail if access is blocked, compatible parts are unavailable, installation detailing prevents removal, actors do not take responsibility, or clients prefer new products before technical failure. The framework therefore checks both product-level and system-level conditions.
Eight-question structure	The framework is organised around eight screening questions. These ask whether the required function is preserved, which parts fail first, whether short-life parts can be replaced independently, whether repair is practically enabled, which actors must support the strategy, what trade-offs are introduced, whether rebound or displacement may occur, and whether the lifecycle benefit remains plausible under transparent assumptions.
How it was trialled	The framework was trialled on the VELFAC 200 ENERGY top-guided residential window case. The trial showed that product documentation, maintenance guidance and component-level repair possibilities provide a basis for product-life extension, but important uncertainties remain around access, installation method, long-term spare-part availability, custom dimensions, end-of-life route and quantified environmental benefit.
Main result of trial application	The trial application suggests that component repair and replacement may be a plausible life-extension strategy for the selected window system, but only under specific enabling conditions. These include accessible components, compatible spare parts, maintenance planning, future upgrade compatibility, responsible actors and verified lifecycle assumptions.
How to interpret the framework	The framework should be used as an early screening tool rather than a final assessment method. It does not calculate environmental impact values and does not replace a complete LCA. Instead, it identifies which assumptions, design decisions and system conditions must be checked before claiming that a circular window strategy produces environmental benefit.
Contribution to the research question	The framework supports the research question by showing how repair, component replacement and product-life extension can be evaluated without assuming that circular strategies are automatically beneficial. It helps identify the conditions under which expected benefits are likely to hold, and the conditions under which rebound risks, implementation gaps, trade-offs or burden shifting may appear.
Limitations	The framework is preliminary and based on a rapid literature review, one product case, two exploratory expert interviews and qualitative lifecycle scenario assumptions. It identifies plausible mechanisms rather than measured effects. Its value is therefore practical and analytical, not predictive or statistically representative.
Final report wording	The Circular Window Component and Rebound-Risk Check provides a practical way to screen repair and component-replacement strategies before they are presented as environmentally beneficial. By combining component-level questions with system-level conditions, the framework helps reveal whether product-life extension is practically enabled, whether benefits depend on uncertain assumptions, and whether unintended consequences should be classified as rebound effects, implementation gaps, lifecycle trade-offs, burden shifting or technological lock-in.